Arundhati Vasishth

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Education

University of Massachusetts, Amherst — Amherst, MA
BS, Computer Science; GPA: 3.83; Chancellor's Award, Dean's List Honoree — *Expected Fall 2028*

Skills

Languages: Python, Java, C, TypeScript, JavaScript, SQL, HTML/CSS
Frameworks & Cloud: React.js, Node.js, Express, Flask, TensorFlow/Keras, AWS (Bedrock, Lambda, DynamoDB, OpenSearch), MongoDB, PostgreSQL
AI & Tools: RAG, LLM integration, Qdrant, MediaPipe, Grok, XRPL, Solana/Metaplex, REST APIs, Git

Experience

Massachusetts Executive Office of Health & Human Services (EOHHS) — Boston, MA
AI Software Engineer — *Spring to Summer 2026*

- Built an AI-powered regulatory chatbot for Massachusetts DPH's statewide Metrik platform, targeting 85%+ accuracy across 351 local health departments and 5 inspection domains.
- Architected an AWS Bedrock RAG pipeline grounded in official Massachusetts CMR documents, delivering near-instant regulatory answers while reducing inspectors' 2–10 minute manual document searches.
- Designed statewide audit logging, inspector feedback, and response-flagging workflows to track accuracy, completeness, citations, and retrieval quality across inspector interactions.

Indian Institute of Science (IISc) — Bangalore, India
Machine Learning Research Intern — *Summer 2025*

- Evaluated 6 ML/DL pipelines across 8,700+ chest X-ray and brain MRI images, implementing SVM, CNN, XGBoost, PCA hybrids, and VGG16 transfer learning for medical image classification.
- Achieved up to 96% test accuracy on brain MRI and 95% on chest X-rays; evaluated accuracy, F1, ROC-AUC, and log-loss across multiple splits while collaborating with 11 researchers.

Minutemen Alternative Investment Fund (MAIF) — Amherst, MA
Quant Arm, Junior Analyst — *Fall 2025 to Present*

- **Neural SDE Volatility Research:** Explored neural stochastic differential equations for implied-volatility modeling and options strategy research through Python/Jupyter notebooks.
- **Prediction-Market Arbitrage:** Built a multi-agent NLP pipeline across 4 exchanges using Qdrant embeddings, KNN clustering, and Groq/Llama classification of 5 semantic relationships; implemented live quote polling and profit-threshold alerts.
- **Earnings Propagation (Ongoing):** Investigating SEC EDGAR filing timestamps and building a point-in-time data pipeline to study earnings-surprise spillovers across company networks; planning XGBoost vs. GCN/GAT evaluation.

Projects

whyrouted | Distributed LLM Load Balancer — Ongoing

- Building a TypeScript load balancer for routing LLM inference traffic across distributed model replicas, implementing 3 scheduling algorithms with 500ms health monitoring to dynamically select healthy, lightly loaded instances.
- Engineering fault-tolerant request handling with automatic replica ejection and 3-attempt failover, achieving <1s failure detection and zero request loss during single-replica failures; validating reliability across 177 automated tests.

Tokenwise | Grid-Aware Multi-Agent LLM Routing System — Fall 2026

- developed a grid-aware LLM router classifying prompts into 5 complexity tiers across 9 local and AWS Bedrock models, achieving 88.7% within-one-tier accuracy on 1,794 labeled prompts.
- Built an AWS Bedrock client and power-grid replay pipeline using historical EIA data; engineered a utility agent negotiating 25% load reductions during peak electricity demand.
- Developed a 351-town crowd simulator, live 3D visualization, and Chrome extension tracking model routing, grid conditions, and energy consumption per response.

KnowYork | Autonomous AI Agent & Blockchain Platform — Fall 2026

- Built a full-stack AI agent that verifies real-world NYC visits and autonomously executes RLUSD rewards across XRPL and Solana, combining Grok vision + multi-sample GPS verification to achieve 90%+ validation accuracy with <4s median decision latency.
- Engineered a zero-trust transaction pipeline with dual-agent verification, deterministic payout controls, and isolated wallet keys, rejecting 95%+ of simulated fraudulent submissions while enforcing a 10 RLUSD on-chain maximum exposure across 25 adversarial security tests.

FingerSolve | AI-Powered Hand Gesture Recognition Platform — Spring 2025

- Built a real-time React/Flask gesture-recognition platform for touch-free math quizzes, improving engagement by 40% and reducing user input errors by 30% during pilot testing.
- Engineered MediaPipe Hands + TensorFlow/Keras classification across 12 gestures with 94% accuracy; deployed a prototype that earned 2nd place at the MassAI ML Conference and attracted interest from 50+ educators.